

Will Ruff

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EXPERIENCE

Research Assistant I

Sep. 2025 – Present

SDSU Climate Informatics Laboratory

San Diego, CA

- Built Python pipelines to process NOAA climate data from AWS S3 into cloud-optimized formats for rapid access
- Leveraged NetCDF, Kerchunk, Zarr, and xarray to build a lightweight, server-side retrieval system for visualization
- Extracted key dataset attributes into a unified metadata layer to make data access more modular and efficient
- Developed FastAPI endpoints to dynamically query and stream climate data from cloud storage on user requests
- Designed an engine-specific access system to automatically handle multiple dataset formats at system runtime

Founding Engineer

Dec. 2025 – Feb. 2026

ToneAdapt.com

San Diego, CA

- Built and scaled a full-stack SaaS to 30,000+ user signups and \$5K+ MRR in the first 60 days
- Implemented a GenAI-assisted research system that aggregates guitar tone data and adapts it to the user's gear
- Architected a normalized database schema and query layer supporting authentication, subscriptions, and usage
- Deployed and maintained production infrastructure on Vercel with automated builds, isolation, and rollbacks
- Developed internal dashboards to track sales, MRR, churn, and user activity for data-driven product decisions

Data Engineer Intern

May 2025 – Aug. 2025

Kenworth - Division of PACCAR Inc.

Seattle, WA

- Engineered an in-house computer vision system replacing a third-party tool, reducing costs by \$100K+ annually
- Improved cab part detection accuracy from 94.5% to 96.5% using supervised learning techniques on annotated data
- Developed custom YOLO and Azure CV models for part detection and cab positioning along the production line
- Built a Flask web application to deliver real-time inference to production dashboards, aiding operator inspections
- Designed a YOLO camera-triggering model to detect optimal cab alignment for capturing all four sides at EOL
- Created a real-time part detection stream with live bounding boxes on cabs as they exit the manufacturing line

Software Engineer Intern

June 2024 – Aug. 2024

Glympse - Location Sharing Technology

Seattle, WA

- Developed Python and Go scripts for file comparison, cutting memory usage by 25% via benchmarking
- Migrated legacy code to a new repository, adding essential parameters and enhancing report functionality
- Refactored and modernized a shell script report in Python, enhancing readability and maintainability
- Built and implemented comprehensive unit tests to ensure the reliability and accuracy of the Python code
- Deployed code using GitLab CI/CD pipelines, configuring environments with '.gitlab-ci.yml' and Docker

PROJECTS

Fitness Pose Detection IOS App | [FlexVault.ai](https://flexvault.ai) | Core ML, Swift, Python

Jul. 2025 – Aug. 2025

- Engineered a custom object detection model with Core ML to classify 8 bodybuilding poses with high accuracy
- Integrated Core ML into an iOS app for on-device inference and real-time pose recognition during photo capture
- Automated web scraping to collect a large dataset of bodybuilding pose photos for annotation and model training
- Implemented side-by-side comparisons, secure storage, and bulk imports to improve organized long-term tracking
- Published to the Apple App Store, releasing a production-ready AI app improving photo tracking for bodybuilders

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, Java, JavaScript, HTML/CSS, FastAPI, Flask

ML/AI: PyTorch, Core ML, YOLO, OpenCV, Object Detection, Azure Custom Vision

Tools/Cloud/DB: AWS (S3), PostgreSQL, Git, Docker, CI/CD Pipelines, Azure Data Factory, Linux, Jupyter

Relevant Coursework: Machine Learning, Algorithms, Artificial Intelligence, Database Systems, Data Science

EDUCATION

San Diego State University

San Diego, CA

Bachelor of Science in Computer Science, Minor in Mathematics

Aug. 2022 – May 2026

Cumulative GPA: 3.9